

Attention Mechanism

A concise technical reference for Transformer-based NLP systems

1. Overview

Attention allows a model to dynamically determine which tokens are most relevant when producing a representation for a particular token or output.

2. Core Components

The mechanism uses three learned projections: Query (Q), Key (K), and Value (V). Similarity between queries and keys determines how strongly each value contributes to the output.

3. Scaled Dot-Product Attention

$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$. Scaling by $\sqrt{d_k}$ helps keep dot products in a numerically stable range before the softmax operation.

4. Multi-Head Attention

Several attention heads operate in parallel. Each head can learn different relationships, such as local context, long-range dependencies, or syntactic associations. Their outputs are concatenated and projected.

5. Applications

Attention is central to modern language models, machine translation, document retrieval, summarization, question answering, and multimodal architectures.